

GB-65-NUTRIENTS

1.SMOTE

Average Accuracy Val K-Fold: 88.8041

Average Accuracy Train K-Fold: 98.1764

Average MCC Val K-Fold: 0.7702

Average MCC Train K-Fold: 0.9757

Average F1 Score Val K-Fold: 0.8920

Average F1 Score Train K-Fold: 0.9817

Average Precision Val K-Fold: 0.9003

Average Precision Train K-Fold: 0.9818

Average Recall Val K-Fold: 0.8880

Average Recall Train K-Fold: 0.9818

Accuracy Test: 88.22%

MCC Test: 0.7575

F1 Score Test: 0.8856

Precision Test: 0.8927

Recall Test: 0.8822

Classification Report:

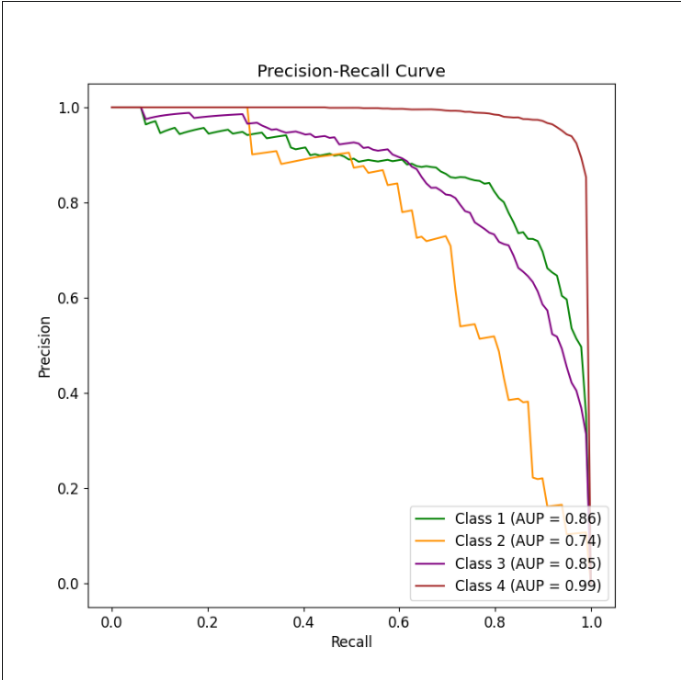
	precision	recall	f1-score	support
1	0.73	0.88	0.80	68
2	0.56	0.82	0.67	11
3	0.72	0.75	0.74	93
4	0.96	0.91	0.94	422
accuracy			0.88	594
macro avg	0.75	0.84	0.79	594
weighted avg	0.89	0.88	0.89	594

Confusion Matrix Test:

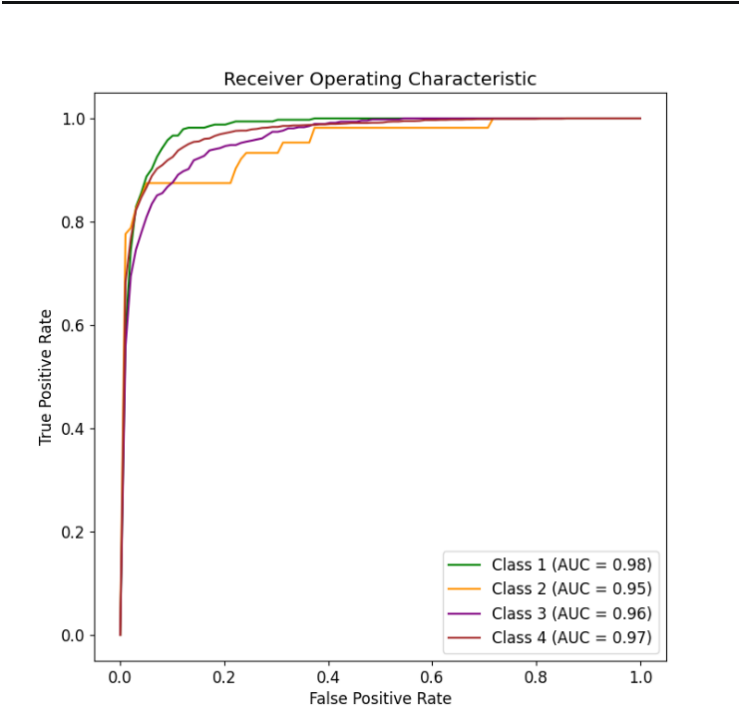
[[60 0 5 3]

[0 9 0 2]
[13 1 70 9]
[9 6 22 385]]

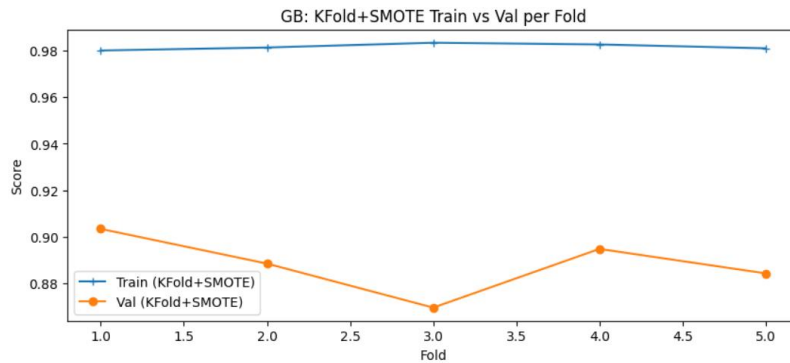
PR curve



ROC curve



Training and Validation curve



2.SMOTE+STRATIFIED

Average Accuracy Val K-Fold: 88.8460

Average Accuracy Train K-Fold: 98.2064

Average MCC Val K-Fold: 0.7737

Average MCC Train K-Fold: 0.9761

Average F1 Score Val K-Fold: 0.8922

Average F1 Score Train K-Fold: 0.9820

Average Precision Val K-Fold: 0.9022

Average Precision Train K-Fold: 0.9821

Average Recall Val K-Fold: 0.8885

Average Recall Train K-Fold: 0.9821

Accuracy Test: 88.55%

MCC Test: 0.7612

F1 Score Test: 0.8884

Precision Test: 0.8931

Recall Test: 0.8855

Classification Report:

	precision	recall	f1-score	support
1	0.74	0.84	0.79	68
2	0.75	0.82	0.78	11
3	0.71	0.77	0.74	93

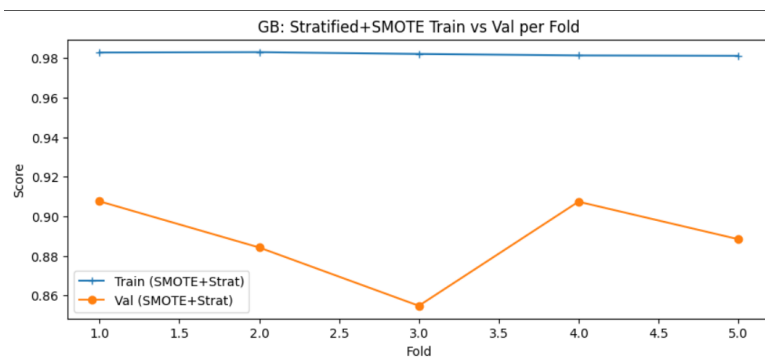
4	0.96	0.92	0.94	422
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accuracy			0.89	594
macro avg	0.79	0.84	0.81	594
weighted avg	0.89	0.89	0.89	594

Confusion Matrix Test:

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[[ 57  0  8  3]
 [  0  9  0  2]
 [ 11  0 72 10]
 [  9  3 22 388]]
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Training and Validation curve



3.STRATIFIED

Average Accuracy Val K-Fold: 90.1934

Average Accuracy Train K-Fold: 98.6322

Average MCC Val K-Fold: 0.7803

Average MCC Train K-Fold: 0.9699

Average F1 Score Val K-Fold: 0.8996

Average F1 Score Train K-Fold: 0.9863

Average Precision Val K-Fold: 0.9006

Average Precision Train K-Fold: 0.9863

Average Recall Val K-Fold: 0.9019

Average Recall Train K-Fold: 0.9863

Accuracy Test: 89.90%

MCC Test: 0.7721

F1 Score Test: 0.8958

Precision Test: 0.8953

Recall Test: 0.8990

Classification Report:

	precision	recall	f1-score	support
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1	0.86	0.72	0.78	68
2	0.75	0.82	0.78	11
3	0.76	0.69	0.72	93
4	0.93	0.98	0.95	422

accuracy			0.90	594
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macro avg	0.83	0.80	0.81	594
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weighted avg	0.90	0.90	0.90	594
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Confusion Matrix Test:

[[49 0 14 5]

[0 9 0 2]

[6 1 64 22]

[2 2 6 412]]

Training and Validation curve

